RENATO SPRENGER-VUKOVIĆ

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SUMMARY

PhD candidate in Computer Science researching large language models, information extraction, and knowledge structuring. Author of multiple peer-reviewed publications at top-tier venues and experienced in teaching and supervising undergraduate and graduate research. Furthermore, I have experience in applying my research to real-world problems through freelance projects in industry, demonstrating the practical impact of my work.

EDUCATION

PhD in Computer Science, Heinrich Heine University — 2023 – 2027 (expected)
Large Language Models, Information Extraction, Task-oriented Dialogue

Master of Computer Science, Heinrich Heine University — 2019 – 2022

Bachelor of Computer Science, Minor in Physics, Heinrich Heine University — 2015 – 2019

EXPERIENCE

Doctoral Researcher — Jan 2023 – Jan 2027
Heinrich Heine University Düsseldorf — *Düsseldorf, Germany*

- Conducted doctoral research on large language models and information extraction, improving ontology construction accuracy by 50% by developing a novel text-to-SQL approach for LLMs.
- Supervised and taught Bachelor's and Master's thesis projects and courses, respectively.

Freelance Machine Learning Engineer — Feb 2024 – Nov 2024
Deutscher Versand Service — *Remote*

- Designed and deployed a GDPR-compliant LLM-based customer service system deployed on Azure, reducing manual workload.

Freelance Data Analyst — Jan 2025 – May 2025
Westdeutscher Rundfunk — *Remote*

- Developed a statistical filtering pipeline to detect anomalous streaming activity, reducing daily variance by up to 50%.

SELECTED PUBLICATIONS

Text-to-SQL Task-oriented Dialogue Ontology Construction - Vukovic et al. - TACL 2026
Post-Training Large Language Models via Reinforcement Learning from Self-Feedback - Van Niekerk et al. - Preprint arXiv 2025
Dialogue Ontology Relation Extraction via Constrained Chain-of-Thought Decoding - Vukovic et al. - SIGDIAL 2024
Less is More: Local Intrinsic Dimensions of Contextual Language Models - Ruppik et al. - NeurIPS 2025
Dialogue Term Extraction using Transfer Learning and Topological Data Analysis - Vukovic et al. - SIGDIAL 2022

SKILLS

Programming	Python (advanced), Java, C++
Machine Learning	PyTorch, HuggingFace, structured prediction, model evaluation, fine-tuning, prompting
Mathematics	Probability, statistics, linear algebra, hypothesis testing, optimization
Software Engineering	Object-oriented design, test-driven development, Git
Cloud	Google Cloud Platform, Microsoft Azure